



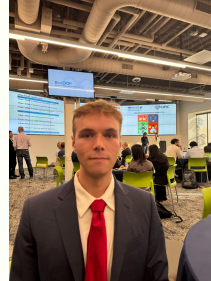
"Eliminating Driveline Complications by Eliminating the Driveline"



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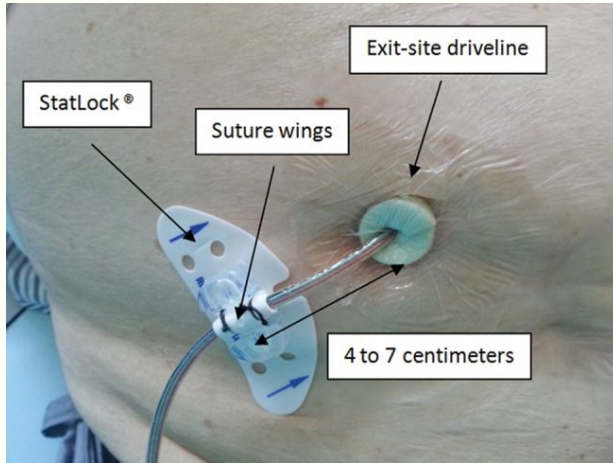
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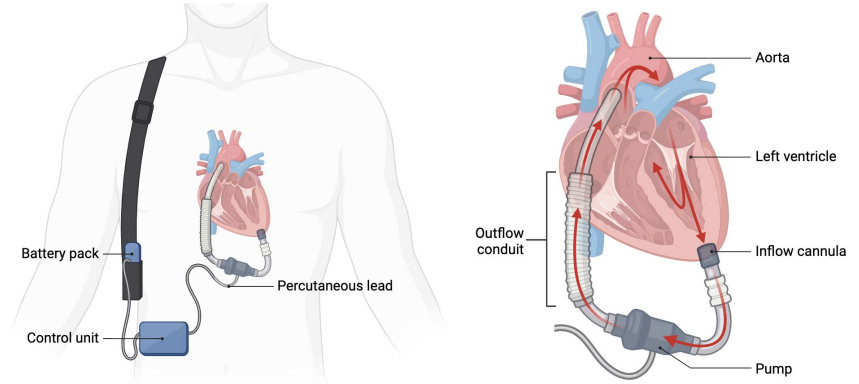


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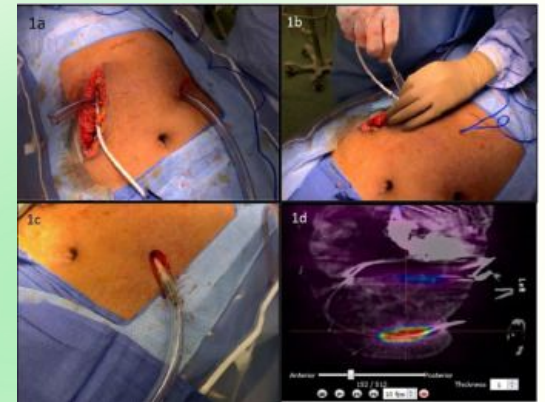


Left Ventricular Assist Device (LVAD)

The LVAD pumps blood into the aorta and then supplied to the body



What are LVADs?



Current Problem

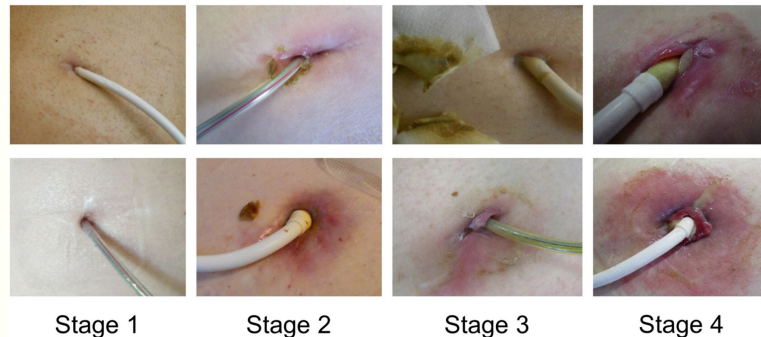
“Drive-line exit-site infections...

occurred in 16.9% (56 of 332) of patients.”

“Sepsis...

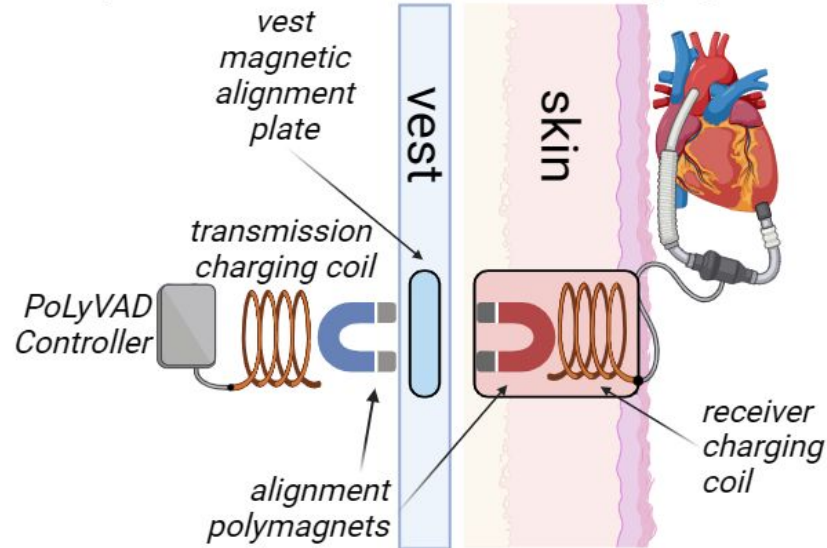
occurred in 17.2% (57 of 332) of patients (0.23 EPPY), and sepsis events occurred more frequently >30 days post-implant (86%, or 60 of 70 sepsis events).”

Examples - modified Sharp Memorial Score (mSC)



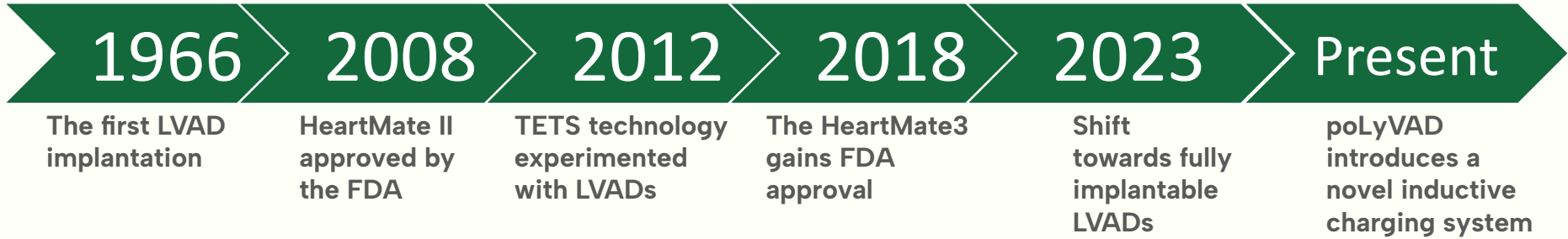
Our Proposed Solution

PoLyVAD Transdermal Inductive Charging Technology



- Polymagnetically-Coupled Inductive Charging System
- Wireless Inductive Power Transfer
- Fully Implantable LVAD eliminates the need for a driveline

Why Now?



Market Trends:

- Advancements in Wireless Energy Transfer
- Rise in Driveline Infection Costs
- Growing Focus on Fully Implantable Medical Devices

Market Size

Value (2020): \$5.7 billion
Projected Value (2025):
\$8.8 billion
CAGR: 9.1%

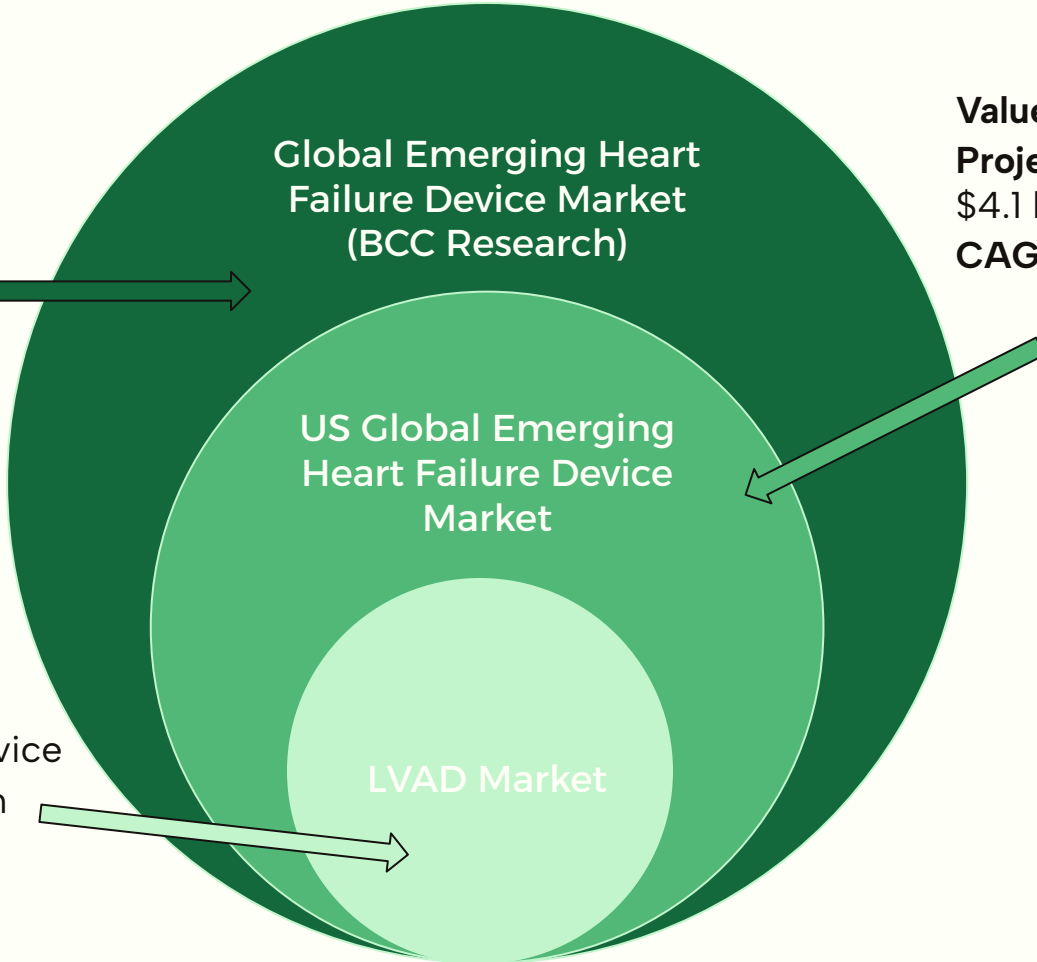
Global Emerging Heart
Failure Device Market
(BCC Research)

Value (2020): \$2.7 billion
Projected Value (2025):
\$4.1 billion
CAGR: 8.3%

US Global Emerging
Heart Failure Device
Market

LVAD Market

Accounts for **20.2%** of
Emerging Heart Failure Device
Value (2020): \$545 million



Competition

Market Landscape

- All presently available LVADs **require drivelines**
- With HVAD (Medtronic) discontinued, Abbott's **HeartMate3 dominates the market**

Advantages

- Small Market
- Only one product stands in our way
- Charging technology easily integrated with LVAD system

Product Comparison

Solution	14 Hour Battery Life	No Drivelines for Risk of Infection	Targeted Magnetic Technology
poLyVAD	✓	✓	✓
Abbott's Fully Implantable Left Ventricular Assist System (FILVAS)	✓	✓	✗
HeartMate3	✓	✗	✗

Go-to-Market



Two options for market entry:

Raise Awareness

Showcase technology via journal publications, trade shows, and cardiology conferences

Partnership with an existing LVAD manufacturer

- Quick market access
- Lower initial costs
- Limited revenue (royalties or licensing fees)
- Limited control

Direct Sales to hospitals and healthcare providers

- Higher revenue potential
- Full control
- Longer and more involved sales cycle
- Higher upfront costs

Financial Projections: Partnership vs Direct Sales

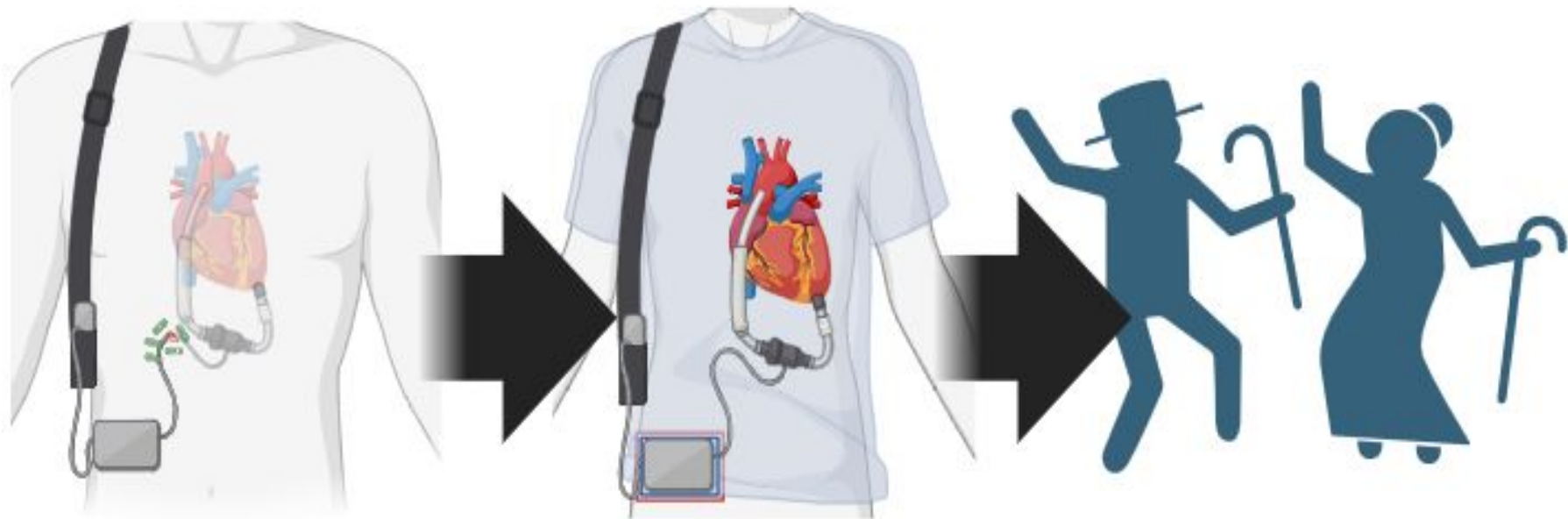
Securing premarket authorization from the FDA will take at least 3-5 years. Once approved, however, **superior LVAD models can quickly take over the LVAD market**

Partnership

- Royalty percentage: **3-8% of gross sales**
- Development and regulatory costs: **\$10 million**
- Assuming speedy market adoption, this venture would break even **soon after FDA approval**

Direct Sales

- Gross profit: **\$50k per LVAD unit**
- Development and regulatory costs: **\$60 million**
- **Revenue and gross margins explode after FDA approval** and increases in production capacity



The Problem:

High Risk of Sepsis and Driveline Infections:

LVADs, which assist in heart function, require a driveline to power the device. This creates an entry point for infection, leading to complications such as sepsis.

Proposed Innovation:

Polymagnetically-Coupled Inductive Charging System:

Replaces the external driveline with a wireless power system that charges the LVAD through the patient's skin.

Expected Outcome/Improvement:

- Reduction in Driveline infections
- Enhanced Patient Comfort and Quality of Life
- Lower Hospital Readmissions





**“Eliminating
Driveline
Complications
by Eliminating
the Driveline”**

Thanks!

Appendix 1 - Patents

Since our proposed innovation is a **functional improvement**, we will apply for a **utility-type patent** and this gives us a **20-year term** from the effective filing date.

As our core inventive concept has already been established and is ready to present, we will take action to file a **provisional/non-provisional application** to the U.S. patent office.

Abbott wins breakthrough tag for fully implantable LVAD system, following Medtronic

<https://www.medtechdive.com/news/abbott-wins-breakthrough-tag-for-fully-implantable-lvad-system-following-m/571670/>

Published Feb. 4, 2020

Stein, Aaron¹; Kovacevic, Milovan¹; Aung Kyaw, Phyo¹; Anderson, Russell²; Verma, Resy²; Provos, John²; Brandmire, Adam²; Iudice, Jeffrey². CARD14: Wireless Transcatheter Energy Transfer and Control System for Powering a Fully-Implanted Left-Ventricular Assist System. ASAIO Journal 69(Supplement 2):p 64, June 2023. | DOI: 10.1097/01.mat.0000943580.83459.85

CARDIAC RESEARCH & DEVELOPMENT ORAL ABSTRACTS

CARD14: Wireless Transcatheter Energy Transfer and Control System for Powering a Fully-Implanted Left-Ventricular Assist System

Stein, Aaron¹; Kovacevic, Milovan¹; Aung Kyaw, Phyo¹; Anderson, Russell²; Verma, Resy²; Provos, John²; Brandmire, Adam²; Iudice, Jeffrey²

Author Information

ASAIO Journal 69(Supplement 2):p 64, June 2023. | DOI: 10.1097/01.mat.0000943580.83459.85

Google Patents

(lvad OR magnet) assignee:(Medtronic, Inc.) assignee:(Abbott Laboratories)

SEARCH TERMS

lvad OR magnet

SEARCH FIELDS

Date - Priority

YYYY-MM-DD - YYYY-MM-DD

Inventor

Medtronic, Inc. x or Abbott Laboratories or Assignee

Patent Office - Language

Status - Type

Litigation

About 29,248 results

Sort by - Relevance - Group by - None - Deduplicate by - Family - Results / page - 10 -

Detection of hypertension in lvad patients using speed change

WO EP US CN JP KR • US20230272136A1 • D'Anne E. Kudlik • Medtronic, Inc.

Priority 2019-02-28 • Filed 2023-05-15 • Published 2023-09-07

used in the MCSO sold under the designation HVAD® by HeartWare, Inc., assignee of the present application. The arrangement of components such as the magnets, electromagnetic coils, and hydrodynamic bearings used in such a pump and variants of the same general design are described in U.S. Pat. Nos.

Corrosion-resistant magnetic article

WO EP US CN JP KR AU GA ES • AU2016209249B2 • Drew Powell Amery • Medtronic Xomed, Inc.

Priority 2015-01-22 • Filed 2016-01-21 • Granted 2021-01-28 • Published 2021-01-28

A magnetic article with a corrosion resistant barrier formed from a poly (tetrafluoro-p- xylene) conformal coating or from a polyene conformal coating having a melting point of at least about 430°C and a moisture vapor transmission less than about 0.5 g-mm/m

Molded vad

WO EP US • US20150151031A1 • Raffy Yaghdjian • Heartware, Inc.

Priority 2013-12-04 • Filed 2014-12-04 • Published 2015-06-04

21. A method of manufacturing a blood pump as claimed in claim 14, further comprising positioning a rotor in the blood-flow lumen, the rotor having a plurality of magnetic poles for interacting with the stator and an impeller with at least one hydrodynamic surface for pumping blood. 22. A method of ...

Device and method for finding the center and reading the setting of an ...

WO EP US CN JP KR AU BR GA ES • US9427559B2 • Deep Shah • Medtronic Xomed, Inc.

Priority 2013-03-14 • Filed 2015-09-08 • Granted 2016-08-30 • Published 2016-08-30

wherein the magnetic capsule is configured to slide within the cavity and center within the target when the tool magnet is magnetically aligned with a magnet of an adjustable valve. 2. The centering tool of claim 1, wherein the housing

Top 1000 results by filing date

Relative count of top 5 values

Assignees	Inventors	CPCs
Medtronic, Inc.		45.9%
Abbott Laboratories		10.8%
Medtronic Minimed, Inc.		5.8%
Biophan Technologies, Inc.		2%
Medtronic Inc		1.8%

Expand

Related landscapes

Appendix 2 - Supporting Infection Info

Complication	HeartMate III	HeartMate II	HeartWare
Sepsis/Driveline Infection	23.6%	15.4-23.6%	15.4%
Stroke	10%	12-20%	25-30%
GI Bleeding event	25%	34-45%	35-39%
Pump Thrombosis	1%	11-12%	6-9%

Figure 2. Chance of each complication happening with each successive device within a 2-3 year implantation.

Appendix 3 - Market Analysis (LVAD Infections)

Hospital Readmissions After Continuous-Flow Left Ventricular Assist Device Implantation: Incidence, Causes, and Cost Analysis

Results: In all, 103 patients underwent implantation for bridge to transplantation and 19 patients for destination therapy; 68 patients were readmitted 156 times (2.2 times/patient) as of the end of follow up. The median follow-up period was 11 months. While receiving device support, patients spent 93% of their time out of the hospital. The causes of readmission included gastrointestinal bleeding (19%), driveline infection (13%), and stroke (8%). The median time to first readmission was 35 days. Thirty (44%) patients were readmitted within 30 days after discharge. The median direct hospital cost of a single readmission was \$7,546. Device malfunction and arrhythmias were the most costly causes of readmission. There was no significant difference in long-term survival between readmitted patients and those who were not readmitted.

Akhter SA, Badami A, Murray M, Kohmoto T, Lozonschi L, Osaki S, Lushaj EB. Hospital Readmissions After Continuous-Flow Left Ventricular Assist Device Implantation: Incidence, Causes, and Cost Analysis. *Ann Thorac Surg*. 2015 Sep;100(3):884-9. doi: 10.1016/j.athoracsur.2015.03.010. Epub 2015 Jun 19. PMID: 26095106.